Method of Characteristics for Minimum Nozzle Length

Nomenclature

Variables

- ν – Prandtl–Meyer angle
- M – Mach number
- μ – Mach angle
- θ – Flow angle

Subscripts

- R – Right Characteristic
- L – Left Characteristic

Method of Characteristics

The method of characteristics (MOC) is a mathematical formulation by considering the fact that PDEs convert to ODEs along the characteristics. In this study, a diverging nozzle contour with minimum length is designed for given throat height, entry and exit Mach numbers. The ratio of specific heats is also to be provided. Characteristics are lines in a supersonic flow oriented in specific directions along which disturbances (pressure waves) are propagated. The three properties of characteristics are Property 1: A characteristic in a two-dimensional supersonic flow is a curve or line along which physical disturbances are propagated at the local speed of sound relative to the gas. Property 2: A characteristic is a curve across which flow properties are continuous, although they may have discontinuous first derivatives, and along which the derivatives are indeterminate. Property 3: A characteristic is a curve along which the governing partial differential equations(s) may be manipulated into an ordinary differential equation(s). The fluid accelerates in the diverging section and in the Prandtl-Meyer expansion scenario, it is assumed that the expansion takes place across a centered fan originating from an abrupt corner .This phenomenon is typically modeled as a continuous series of expansion waves, each turning the airflow an infinitesimal amount along with the contour of the channel wall.

The Prandtl–Meyer angle is calculated by using Prandtl–Meyer function (Eq. 1) and it is function of γ and M. Similarly using ν, Mach number can be calculated using the approximate equation as given in Eq. 2.

$$\nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}} tan^{-1}\left[\frac{\gamma-1}{\gamma+1}(M^2-1)\right] - tan^{-1}(M^2-1) \tag{1}$$

$$M(\nu) = \frac{1 + Ax + Bx^2 + Cx^3}{1 + Dx + Ex^2} \tag{2}$$

$$where, x = \left[\frac{\nu}{\nu_{inf}}\right]^{2/3}, \nu_{inf} = 0.5\pi(\sqrt{6}-1)$$

$$A = 1.3604, B = 0.0962, C = -0.5127, D = -0.6722, E = -0.3278$$

The pictorial representation of left ($\nu - \theta = K_+$) and right ($\nu + \theta = K_-$) characteristics is shown in Fig.1. All the angles are represented in Fig. 2 for better understanding of

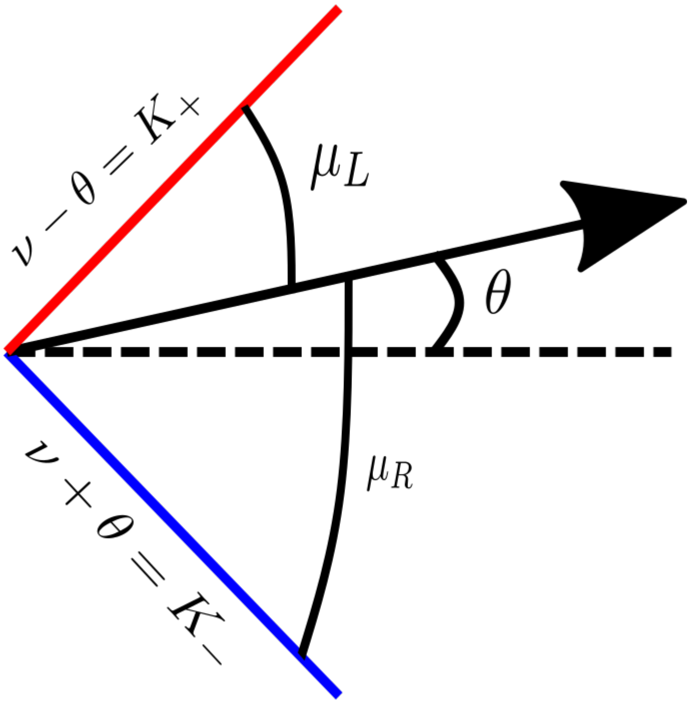

Figure 1: Left and Right characteristics representation

the method. AB is vertical reference and AD is horizontal reference. A1 and AN are the extreme expansion waves of the expansion fan formed at the corner A. This is 2D analysis and because of symmetry only the top half will be shown. Initially the flow direction is along the horizontal reference and after the flow passes the AN expansion wave, the flow

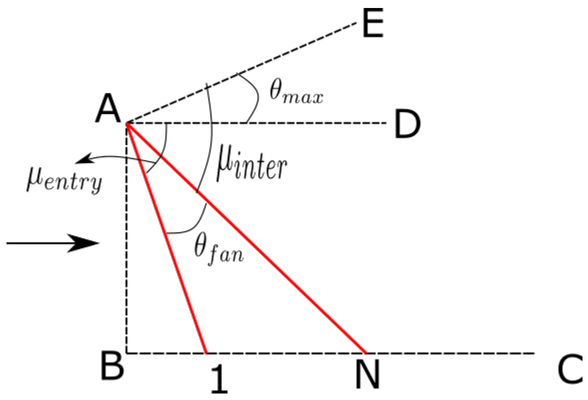

Figure 2: Different angles required for MOC

direction is along AE. The angle at which AE makes with horizontal reference AD is given by Eq. 3. The ν values can be calculated using Eq. 1 with the given entry and exit Mach numbers. Since the expansion waves start from both the corners (A and its reflection w.r.t BC) the flow will be accelerated due to the interaction of expansion waves from both the sides. So the intermediate Mach number right after the AN expansion wave will be based on $\nu_{inter} = \nu_{entry} + \theta_{max}$. From ν_{inter}, the M_{inter} can be calculated using Eq. 2. Once M_{inter} is obtained, we get $\mu_{inter}(= sin^{-1}(\frac{1}{M_{inter}}))$. From the obtained information, we can calculate the expansion fan angle (θ_{fan}) from Eq. 4. Once we get θ_{fan}, we can use any number of expansion waves between A1 and AN by considering $\Delta\theta$.

$$\theta_{max} = \frac{\nu_{exit} - \nu_{entry}}{2} \tag{3}$$

$$\theta_{fan} = \mu_{entry} + \theta_{max} - \mu_{inter} \tag{4}$$

For the first characteristic A1, since it is a right characteristic ($\nu + \theta = K_-$), K_- can be found as we know ν_{entry} and $\theta = 0$ as the flow is along AD for first characteristic (also, $\theta = 0$ for all points that lie on BC). For the second right characterisitic originating from A, $\theta = \Delta\theta$ and $\nu = \nu_{entry} + \Delta\theta$. Therefore, $K_- = \nu_{entry} + 2\Delta\theta$ for second right characteristic. Similarly all the K_- values for the right characteristics originating from A can be found. Since the left characteristics is mirror image of the right characteristics, $K_+ = K_-$ correspondingly. As the left and right characteristics interact with each other, ν and θ at the intersection points can be calculated using the Eq's. 5 and 6.

$$\theta = \frac{K_- + K_+}{2} \tag{5}$$

$$\nu = \frac{K_- - K_+}{2} \tag{6}$$

Once ν and θ values are calculated at all the points, the slopes of the lines joining these points can be calculated using Eq's. 7 and 8. After the slopes are obtained, the co-ordinates of the points can be obtained by considering intersection of lines. The characteristics with intersection points is shown in Fig. 3 along with the final nozzle contour in black colour.

$$m_- = tan(\theta - \mu) \tag{7}$$

$$m_+ = tan(\theta + \mu) \tag{8}$$

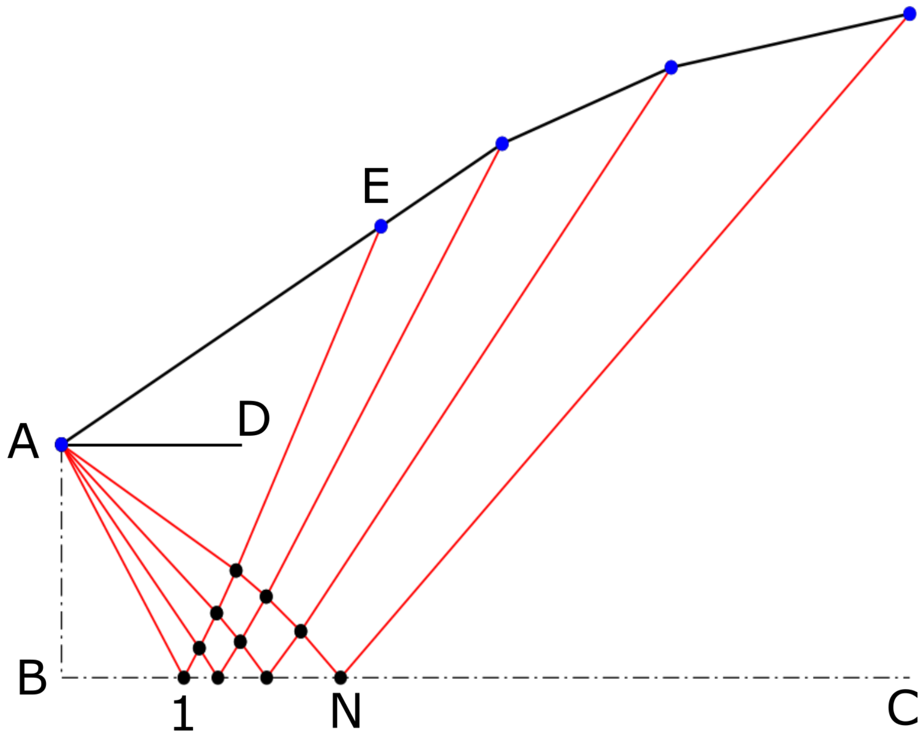

Figure 3: Characteristics and nozzle contour